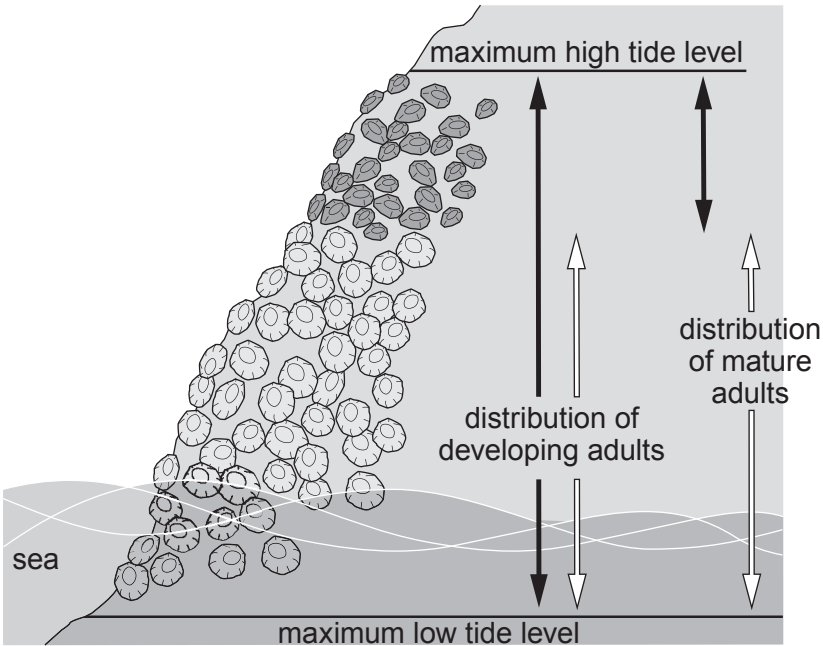


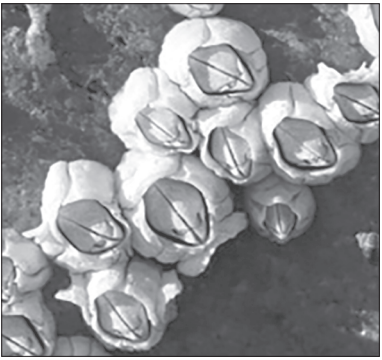
5. Barnacles are marine animals related to crabs and lobsters. They have an external shell made of several plates. They live in shallow tidal waters and the adults are attached permanently to a hard surface. They feed by filtering particles from the water using their modified feathery legs. When they reproduce, they produce larvae which swim in the sea until they attach to a hard surface and begin to grow into adults.

Barnacles are commonly found on rocky shores around the coast of Wales. The two most common species are *Semibalanus balanoides* and *Chthamalus stellatus*.

A rocky shore was studied and the distribution of these species was found as shown in the diagram.



- ↔ Distribution of *Chthamalus stellatus*
- ↔ Distribution of *Semibalanus balanoides*
-  *Chthamalus stellatus*
-  *Semibalanus balanoides*



Chthamalus stellatus



Semibalanus balanoides



Examiner
only

(a) Near the maximum high tide level only *Chthamalus* barnacle larvae develop into adults.

(i) State the type of competition occurring in this area and what the mature adults could be competing for. [2]

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(ii) Scientists removed *Chthamalus* from the upper area continually for several weeks and observed that *Semibalanus* did not colonise this area. Suggest a possible explanation why *Semibalanus* cannot colonise this area. [1]

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(iii) Scientists then removed *Semibalanus* from the lower area continually for several weeks and observed that *Chthamalus* were found in this cleared area. Give an explanation for this observation. [2]

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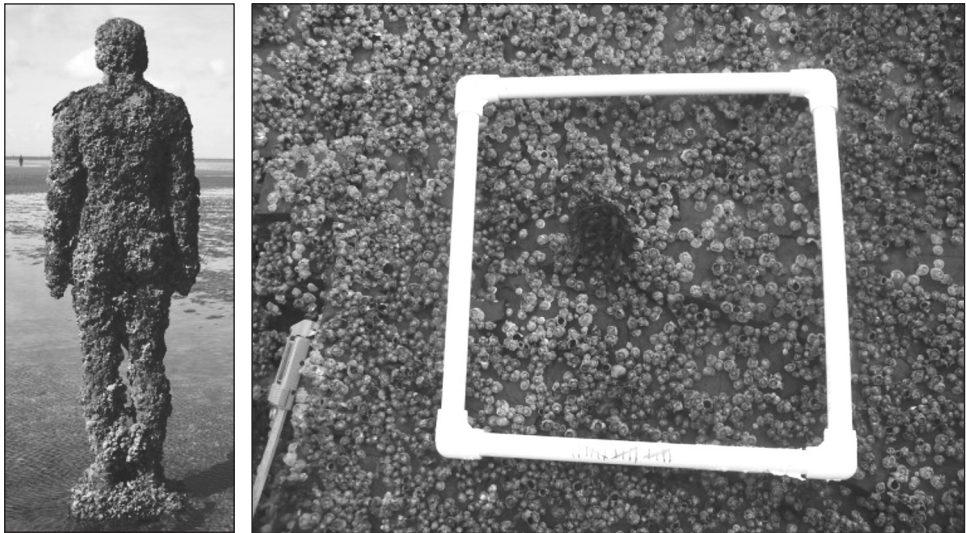
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Examiner
only

In 2005, 100 life-sized human sculptures were installed, at different distances between low and high tide marks, on a sandy beach near Liverpool. A year later it was noticed that some of these structures had become covered in barnacles. All of the barnacles belonged to the species *Austrominius modestus*. The photographs below show one of the statues and a 0.1 m × 0.1 m quadrat being used on the statue.



Scientists selected sculptures to look at the distribution of the barnacles at different positions on the shore.

(b) (i) Describe the method by which reliable data would have been collected. [4]

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Examiner
only

- (ii) If there is decreased salinity in the surrounding water (e.g. in an estuary) *Austrominius* are able to carry out active transport to remove sodium ions from their cells. Explain why they would be unable to survive if this did not occur. [3]

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- (c) Two varieties of *Austrominius* were found on one of the sculptures.

- (i) A sample of DNA was taken from each of the varieties. State the technique that could be used to show the relatedness of the two varieties of barnacles. [1]

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- (ii) State **two** possible causes of increased DNA mutation rate. [1]

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Examiner
only

- (d) *Austrominius modestus* is a species of barnacle that was found only in Australasia up to 100 years ago. The first recorded sighting of it in the UK was in 1946. In 2014, a survey of sixteen harbours in Wales found *Austrominius* in all of the harbours surveyed. It is now classified as an “invasive species”.

At the time of the survey there were just a few *Austrominius* barnacles in each of the harbours.

State why the Welsh Government is concerned about the presence of *Austrominius* in the harbours. Your answer should include environmental and economic concerns. [4]

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